



## AQ8.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

Consider an ordered basis  $\gamma = \{b_1, b_2, b_3\}$  of the inner product space  $R^3$  with respect to the dot product, where  $b_1 = (1, 2, 3)$ ,  $b_2 = (1, 1, 1)$  and  $b_3 = (0, 2, 1)$ . Let  $\beta = \{a_1, a_2, a_3\}$  be the orthonormal basis which is obtained using the Gram-Schmidt process from the basis  $\gamma$ . Use this information to answer the questions 1 and 2.

1 point

The vector  $a_2$  is

☐

$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

☐

$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

☐

$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

☐

$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{-4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{-4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

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$$\left(\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

☐

$$\left(\frac{4}{\sqrt{21}}, \frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}\right)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:



$$\left(\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{-2}{\sqrt{21}}\right)$$

✓

$$4, 21$$

✓

$$1, 21$$

✓

$$-2)$$

1 point

The vector  $a_3$  is

☐

$$\frac{1}{\sqrt{6}}(-1, 2, -1)$$

✓

$$1(-1, 2, -1)$$

☐

$$\frac{1}{\sqrt{6}}(-1, -2, -1)$$

✓

$$1(-1, -2, -1)$$

☐

$$\frac{1}{\sqrt{6}}(1, 2, -1)$$

✓

$$1(1, 2, -1)$$

☐

$$\frac{1}{\sqrt{6}}(1, 2, 1)$$

✓

$$1(1, 2, 1)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{1}{\sqrt{6}}(-1, 2, -1)$$

✓

$$1(-1, 2, -1)$$

## Level 2

1 point

Let  $v_1 = (1, 0, 1, 1)$  and  $v_2 = (0, 1, 1, 1)$  be the vectors from the inner product space  $R^4$  with respect to the dot product. If  $v_3 = v_2 + av_1$  where  $a \in R$  and  $v_1, v_3$  are orthogonal, then

☐

$$a = -2/3$$

☐

$$a = 2/3$$

☐

$$a = 1/3$$

☐

$$a = -1/3$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$a = -2/3a = -2/3$$

1 point

Let  $W$  be a subspace of the inner product space  $R^4$  with respect to dot product and  $\{v_1, v_2, v_3\}$  be an ordered basis of  $W$ , where  $v_1 = (1, 1, 1, 1)$ ,  $v_2 = (1, 1, 1, 0)$ ,  $v_3 = (1, 1, 0, 0)$ . Which of these is the orthonormal basis of  $W$  obtained using the Gram-Schmidt process?

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{\sqrt{6}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, 2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{\sqrt{6}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{\sqrt{6}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{\sqrt{6}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, 2, -2) \right\}$$

**No, the answer is incorrect.****Score: 0****Accepted Answers:**

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{\sqrt{6}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(1, 1, -2, 0) \right\}$$

☐

$$\left\{ \frac{1}{2}(1, 1, 1, 1), \frac{1}{2\sqrt{3}}(1, 1, 1, -3), \frac{1}{\sqrt{6}}(3, 1, -2, 0) \right\}$$

1 point

Let  $a = (\frac{2}{3}, \frac{2}{3}, \frac{1}{3})$  be a vector from the inner product space  $R^3$  with respect to dot product and  $W = \{(x, y, z) \in R^3 \mid \langle (x, y, z), (\frac{2}{3}, \frac{2}{3}, \frac{1}{3}) \rangle = 0\}$  be a subspace of  $R^3$ . Then which of the following is (are) a basis of  $W$ ?

☐

$$\{(1, 0, 2), (0, 1, 2)\}$$

☐

$$\{(1, 0, -2), (0, 1, -2)\}$$

☐

$$\{(-1, 0, 2), (0, -1, 2)\}$$



$\{(1, 0, 2), (0, 1, -2)\} \{(1, 0, 2), (0, 1, -2)\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(1, 0, -2), (0, 1, -2)\} \{(1, 0, -2), (0, 1, -2)\}$

$\{(-1, 0, 2), (0, -1, 2)\} \{(-1, 0, 2), (0, -1, 2)\}$

[Check Answers](#)

Your score is: 0/5